Section A: Multiple Choice Questions (15 marks)

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct and indicate your answers on the separate answer sheet provided.

1 A mixture containing H_2S and CS_2 in a 2 : 3 ratio was burnt in excess oxygen.

What will be the $SO_2 : CO_2 : H_2O$ ratio in the mixture obtained after complete combustion?

A	3 : 1 : 1	**B**	7 : 2 : 3	**C**	8 : 3 : 2	**D**	8 : 3 : 4

Answer: **C**

$2H_2S + 3O_2 \rightarrow 2H_2O + 2SO_2$

$3CS_2 + 9O_2 \rightarrow 3CO_2 + 6SO_2$

8SO_2 : **3**CO_2 : **2**H_2O

2 Cl_2 disproportionates completely in alkaline medium to give 5.0×10^{-3} mol of Cl^- and 1.0×10^{-3} mol of ClO_x^-.

What is the value of x?

A	4	**B**	3
C	2	**D**	1

Answer: **B**

$Cl_2 + e \rightarrow Cl^-$

Based on the above mol ratio, Cl_2 gains 5.0×10^{-3} mol of e^- to form 5.0×10^{-3} mol of Cl^-

Hence Cl_2 gains 5.0×10^{-3} mol of e^- to form 1.0×10^{-3} mol of ClO_x^-

Thus the oxidation state of Cl in ClO_x^- is +5.

$+5 + x(-2) = -1$

$x = 3$

3 *Use of the Data Booklet is relevant in this question.*

When nuclear reactions take place, the elements produced are different from the elements that reacted. An example of a nuclear reaction is given below:

$$^{235}_{92}U + ^{1}_{0}n \longrightarrow ^{144}_{56}Ba + ^{89}_{36}Kr + 3\ ^{1}_{0}n$$

In another nuclear reaction, uranium-238 reacted with deuterium atoms, $^{2}_{1}H$ to form an isotope of a new element, **Q**, as well as two neutrons.

$$^{238}_{92}U + ^{2}_{1}H \longrightarrow \mathbf{Q} + 2\ ^{1}_{0}n$$

	What is isotope **Q**?

	A	^{238}Pu	**B**	^{238}Np	**C**	^{240}Pu	**D**	^{240}Np

The sum of the mass number of the reactants = the sum of the mass number of the products.

Mass number of **Q** = (238 + 2) – 2
= 238

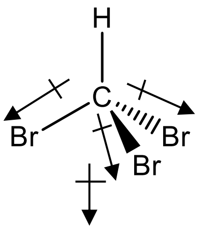
Likewise, the sum of the proton number of the reactants = the sum of the proton number of the products.

Proton number of **Q** = (92 + 1) – 0
= 93

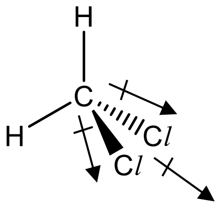
Hence, from the periodic table, **Q** is $^{238}_{93}Np$. (Option B)

4	Which pair of molecules is polar?

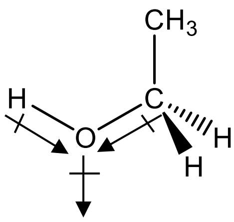
A	$CHBr_3$	CCl_4
B	C_2Br_4	CH_2Cl_2
C	$CHBr_3$	CH_2Cl_2
D	C_2Br_4	C_2H_5OH



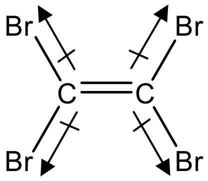
net dipole moment (polar)



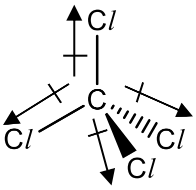
net dipole moment (polar)



net dipole moment (polar)



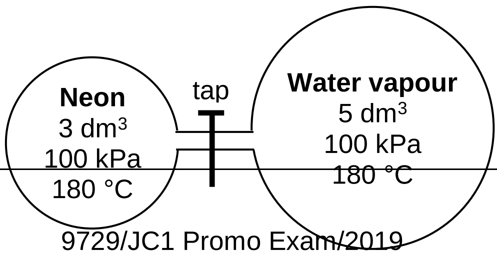
no net dipole moment (non-polar)



no net dipole moment (non-polar)

(Option C)

5	Two bulbs are connected as shown in the diagram below. The bulbs are connected by a narrow tube of negligible volume.



	When the tap is opened, the two gases mix. The connected bulbs were then allowed to cool to room temperature. What was the final pressure, in kPa, in the connected bulbs?

	A	13.9	**B**	24.3	**C**	37.5	**D**	64.7

When the connected bulbs were allowed to cool to room temperature, water vapour condensed to give liquid water, which occupy negligible volume. Hence, we only need to consider the amount of neon gas present in the connected bulbs.

PV = nRT
$100000 \times (3 \times 10^{-3}) = n \times 8.31 \times (180 + 273)$
$n = 0.07969$ mol (amount of neon)

When the tap is opened, and the total volume is 8 dm^3,

PV = nRT
$P \times (8 \times 10^{-3}) = 0.07969 \times 8.31 \times 293$
$P = 24254$ Pa
$= 24.3$ kPa (Option B)

6	The enthalpy change of formation of water and carbon dioxide is –286 kJ mol^{-1} and –394 kJ mol^{-1} respectively. What is the ratio of heat energy generated by hydrogen to that of carbon if the same mass of each substance is burned in excess oxygen?

	A	0.7	**B**	1.4	**C**	4.4	**D**	8.7

$H_2 (g) + 1/2\, O_2 (g) \longrightarrow H_2O (l)$ $\Delta H^{\ominus}_{formation} (H_2O) = \Delta H^{\ominus}_{combustion} (H_2) = –286$ kJ mol^{-1}

$C (s) + O_2 (g) \longrightarrow CO_2(g)$ $\Delta H^{\ominus}_{formation} (CO_2) = \Delta H^{\ominus}_{combustion} (C) = –394$ kJ mol^{-1}

Let the mass of hydrogen and carbon be y respectively.

Amount of H_2 = y/2 mol
Heat released by combustion of H_2 = (y/2) x 286 = 143y kJ

Amount of C = y/12 mol
Heat released by combustion of C = (y/12) x 394 = 32.83y kJ

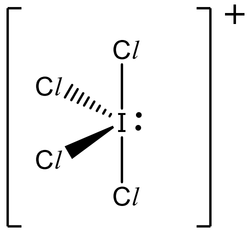
Ratio of heat energy generated by hydrogen to that of carbon = 143y / 32.83 y
= 4.4 (Option C)

7	The shapes of two species **X** and **Y** are see-saw and square planar respectively. Which can be **X** and **Y**?

	X	**Y**
A	ICl_4^+	ICl_4^-
B	ICl_4^-	XeF_4

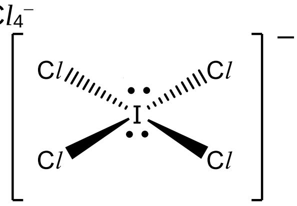
	C	AlH_4^-	ICl_4^+
	D	SF_4	AlH_4^-

ICl_4^+



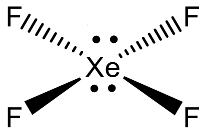
see-saw

ICl_4^-



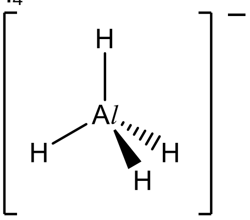
square planar

XeF_4



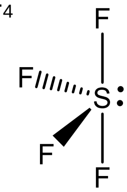
square planar

AlH_4^-



tetrahedral

SF_4



see-saw

(Option A)

8	The numerical values of the equilibrium constant, K_p, for the reaction $$Ag_2CO_3(s) \rightleftharpoons Ag_2O(s) + CO_2(g)$$ are 3.20×10^{-3} and 1.50 at 298 K and 500 K respectively. Which statement is correct?	
	A	The backward reaction is endothermic.
	B	The yield of carbon dioxide increases with temperature.
	C	The value of K_p depends on the amount of Ag_2CO_3 used.
	D	The yield of carbon dioxide will increase when Ag_2O is removed.
	Answer: **B** Since the value of K_p increases, the forward reaction is favoured when temperature increases. The forward reaction is endothermic. (A is incorrect) When temperature increases, forward reaction is favoured, and more CO_2 is formed. (B is correct) Ag_2CO_3 and Ag_2O are solids, so they are not part of the K_p expression, hence their amounts used does not affect the value of K_p or affect the position of equilibrium. (C and D are incorrect)	

9 Which factors affect both the value of K_c and rate of the forward reaction for a reversible reaction?

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta H = \text{negative}$$

1 An increase in temperature

2 An increase in pressure

3 Addition of a catalyst

A 1 only

B 1 and 2 only

C 1 and 3 only

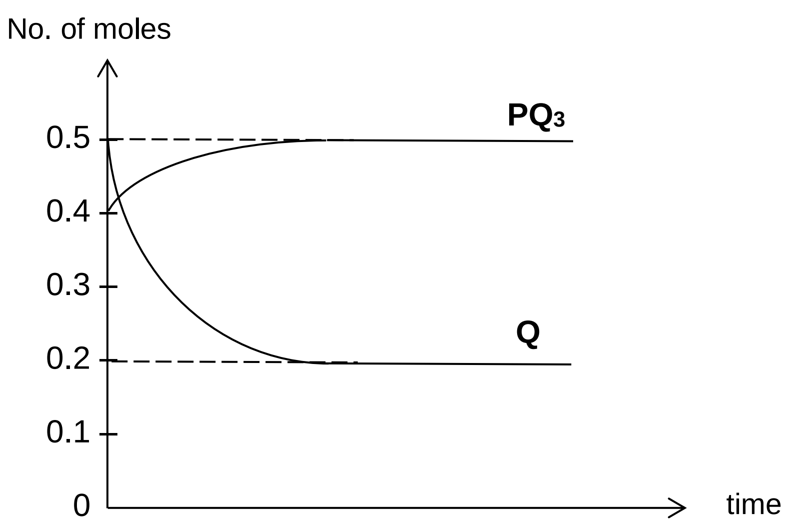
D 2 and 3 only

Answer: **A**
Pressure only affects the rate of forward reaction for gaseous reaction.
Catalyst only increases the rate of forward and backward reactions.

10 The diagram below shows the change in number of moles of $\mathbf{PQ_3}$ and **Q** with time. The initial number of moles of **P** was 0.3.

The system containing **P**, **Q** and $\mathbf{PQ_3}$ is allowed to reach equilibrium in a 5 dm^3 vessel at a temperature of 1000 K.

$$\mathbf{P}(g) + 3\mathbf{Q}(g) \rightleftharpoons \mathbf{PQ_3}(g)$$



Using information from the graph, what is the equilibrium constant K_c for the reaction?

A	$\dfrac{0.5}{0.1 \times (0.2)^3}$	**B**	$\dfrac{0.5}{0.2 \times (0.2)^3}$	**C**	$\dfrac{0.5 \times 5^3}{0.1 \times (0.2)^3}$	**D**	$\dfrac{0.5 \times 5^3}{0.2 \times (0.2)^3}$

Answer: **D**

$K_c = \dfrac{[PQ_3]}{[P][Q]^3}$

Since the volume of the vessel is 5 dm^3,

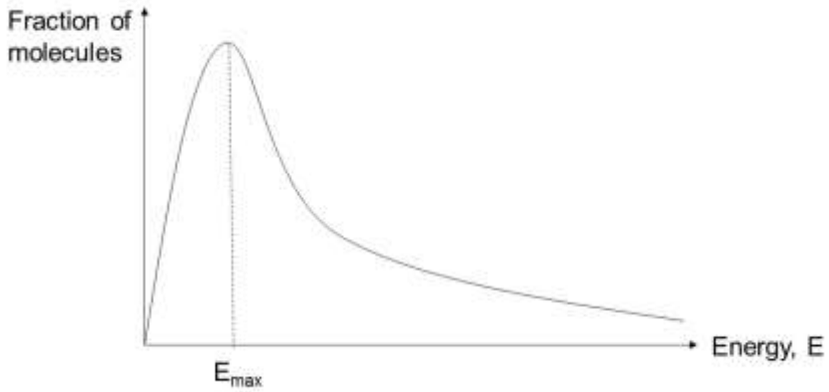
$[PQ_3]_{eq}$ = 0.5 / 5, $[Q]_{eq}$ = 0.2 / 5, $[P]_{eq}$ = $(0.3 - \frac{0.3}{3})$ / 5 = 0.2 / 5

$K_c = \dfrac{(0.5 \div 5)}{(0.2 \div 5)(0.2 \div 5)^3}$

$$= \frac{(0.5)}{(0.2)(0.2 \div 5)^3}$$

$$= \frac{(0.5 \times 5)}{(0.2)(0.2)^3}$$

11 The diagram below represents the Boltzmann distribution of molecular energies at a given temperature.

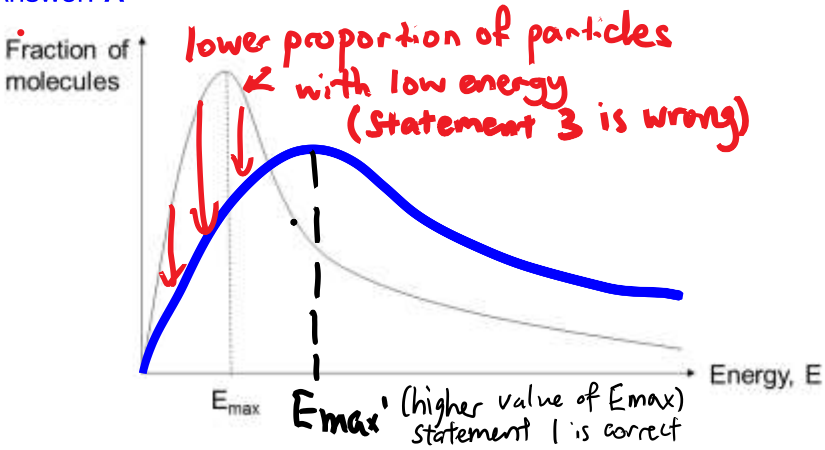


With an increase in temperature, which statements are correct?

1	The value of E_{max} increases.
2	The value of the rate constant, *k*, decreases.
3	At all energies, the proportion of molecules increases.

A	1 only	**B**	1 and 3 only
C	2 and 3 only	**D**	1, 2 and 3

Answer: **A**



By Arrhenius Equation, k = $Ae^{-\frac{Ea}{RT}}$, when temperature increases rate constant k also increases. (Statement 2 is wrong)

12 Which equation represents an *addition* reaction?

A $CH_3CHO + HCN \longrightarrow CH_3CH(OH)CN$

B $CH_3COCl + H_2O \longrightarrow CH_3COOH + HCl$

C $(CH_3)_3CBr + OH^{-} \longrightarrow (CH_3)_3COH + Br^{-}$

D $CH_3CH_2Br \longrightarrow C_2H_4 + HBr$

B Hydrolysis

C Substitution

D Elimination

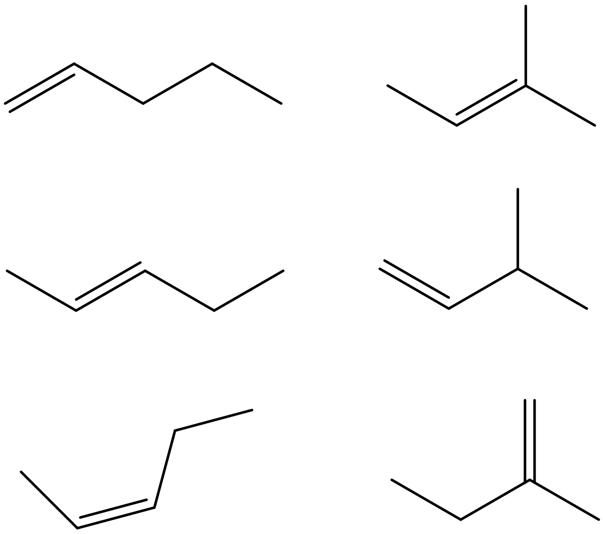
13 What is the number of isomers (including both constitutional and stereoisomers) possible for all open chain hydrocarbons represented by the formula C_5H_{10}?

A 3

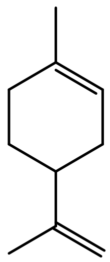
B 4

C 5

D 6



14 Limonene is a major component in the oil of citrus fruit peels.



Limonene

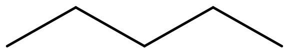
How many σ and π bonds are there in a molecule of limonene?

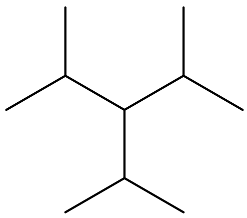
	σ bonds	π bonds

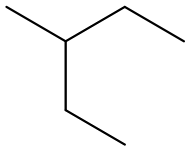
A	10	2
B	26	2
C	10	4
D	26	4

15 One hydrogen atom of hydrocarbons may be substituted with a halogen atom under suitable conditions.

Which hydrocarbons, upon mono-substitution, will give 3 different positional isomers as products?

1 

2 

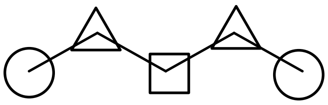
3 

A 1 only

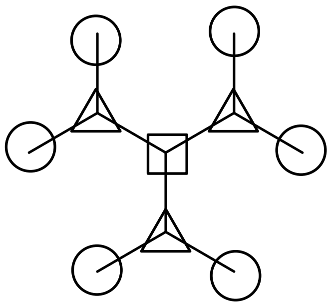
B 3 only

C 1 and 2 only

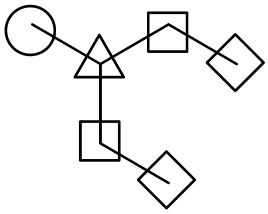
D 1, 2 and 3



3 isomers



3 isomers

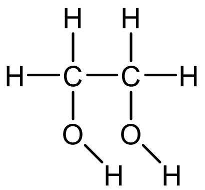


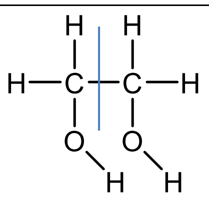
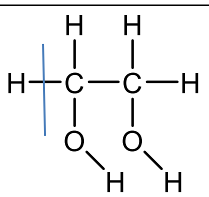
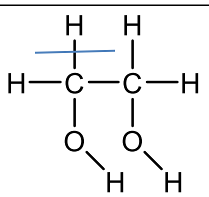
4 isomers

Blank Page

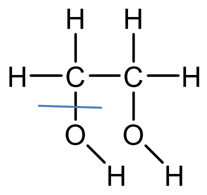
Section B: Structured Questions (30 marks)
Answer **all** the questions in the spaces provided.

For Examiner's Use

1	Sodium bismuthate $NaBiO_3$ oxidises ethane-1,2-diol, $CH_2(OH)CH_2OH$, to **X** under acidic conditions. Sodium bismuthate is reduced to Bi^{3+} during the reaction.	
	(a)	Draw the displayed formula of ethane-1,2-diol. [1]
		
		Comments • The question asked for a displayed formula, hence all bonds must be shown. • A common mistake was the O–H bond was not shown.
	(b)	State the oxidation number of carbon in ethane-1,2-diol. [1]
		–1
		Comments Only one C needs to be considered since the molecule is symmetrical. Consider what happens to the shared electrons when each bond is broken.

Bond broken	Where electron goes	Effect on oxidation state on C
	When C-C bond is broken, both C has equal electronegativity, thus each C will take back their own electron.	0
	When C-H bond is broken, C is more electronegative than H. Both electrons in C-H bond goes to C. Thus, C will take back its own electron and gain one electron originally contributed by H	C gained one electron, thus will be –1
	When C-H bond is broken, C is more electronegative than H. Both electrons in C-H bond goes to C. Thus, C will take back its own electron and gain one electron originally contributed by H	C gained one electron, thus will be –1

For Examiner's Use

	When C-O bond is broken, O is more electronegative than C. Both electrons in C-O bond goes to O. Thus, C will lose its own electron to O.	C lost one electron, thus will be +1

Thus, net oxidation state of C after all 4 bonds are broken = 0 – 1 – 1 + 1
= –1

- Note that for oxidation state, the sign should be written in front of the number.
- 1– is not acceptable as an answer.

(c) Write a balanced half - equation for the reduction of BiO_3^- to Bi^{3+}. [1]

$BiO_3^- + 6H^+ + 2e^- \rightarrow Bi^{3+} + 3H_2O$

Comments

- The question mentioned that the medium is acidic hence H^+ should be used to balance the H atoms in the equation.

(d) Sodium bismuthate reacts with ethane-1,2-diol in a 1:1 mole ratio.

(i) Deduce the final oxidation number of carbon in **X**. [1]

2 moles of electrons are gained per mole of BiO_3^- and <u>2 moles of electrons are lost per mole of ethan-1,2-diol.</u>

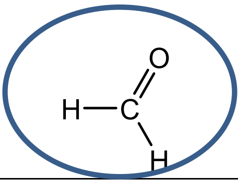
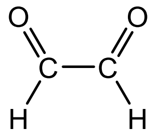
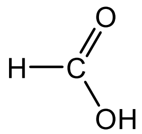
<u>1 mole of electron is lost per carbon atom</u> of ethan-1,2-diol so increase of oxidation number from <u>–1 to 0</u>. (any form of working is ok)

Comments

- Bi in BiO_3^- is reduced since it takes in two electrons on the left.
- Thus, ethane-1,2-diol is oxidised and the C atom should be oxidised.
- A common mistake was not to recognise that there are two carbons in ethane-1,2-diol, hence, the two electrons that Bi in BiO_3^- took up comes from both the C atoms.
- Since two electrons are contributed by two carbon atoms, each carbon atom only gives up one electron.
- Hence, the oxidation state of each carbon atom in ethan-1,2-diol will only increase by one unit, from –1 to 0.

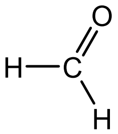
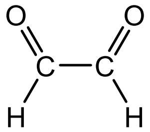
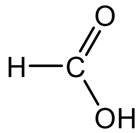
For Examiner's Use

(ii) **X** is one of the three structures shown below.
Circle the correct one. [1]

Methanal

Comments:

		
Break C-H bond: C more electronegative than H so C takes both electrons in the bond. Net gain in one electron for C from H. Two C-H bonds broken so C gains 2 electrons from 2 H atoms. **Break C=O bond:** 4 electrons in the C=O bond. O more electronegative than C so O gains all 4 electrons. 2 electrons are originally contibuted by O and 2 electrons are contributed by C. Thus, C lost 2 electrons to O. **Net:** C gained 2 electrons from H and lost 2 electrons to O **Oxidation state = –2 +2 = 0**	**For each C:** **Break C-H bond:** C more electronegative than H so C takes both electrons in the bond. Net gain in one electron for C from H. **Break C=O bond:** 4 electrons in the C=O bond. O more electronegative than C so O gains all 4 electrons. 2 electrons are originally contibuted by O and 2 electrons are contributed by C. Thus, C lost 2 electrons to O. **Break C-C bond:** Each C gets back its own electron so no change. **Net:** C gained 1 electron from H and lost 2 electrons to O **Oxidation state = –1 +2 = +1**	**Break C-H bond:** C more electronegative than H so C takes both electrons in the bond. Net gain in one electron for C from H. **Break C=O bond:** 4 electrons in the C=O bond. O more electronegative than C so O gains all 4 electrons. 2 electrons are originally contibuted by O and 2 electrons are contributed by C. Thus, C lost 2 electrons to O. **Break C-O bond:** 2 electrons in the C-O bond. O more electronegative than C so O gains both electrons in bond. Thus, C lost 1 electron to O. **Net:** C gained 1 electron from H, lost 2 electrons to O in C=O bond broken, lost 1 electron to O in C-O bond broken. **Oxidation state = –1 +2+1 = +2**

[Total: 5]

For Examiner's Use

2 (a) Complete the electronic configuration of:

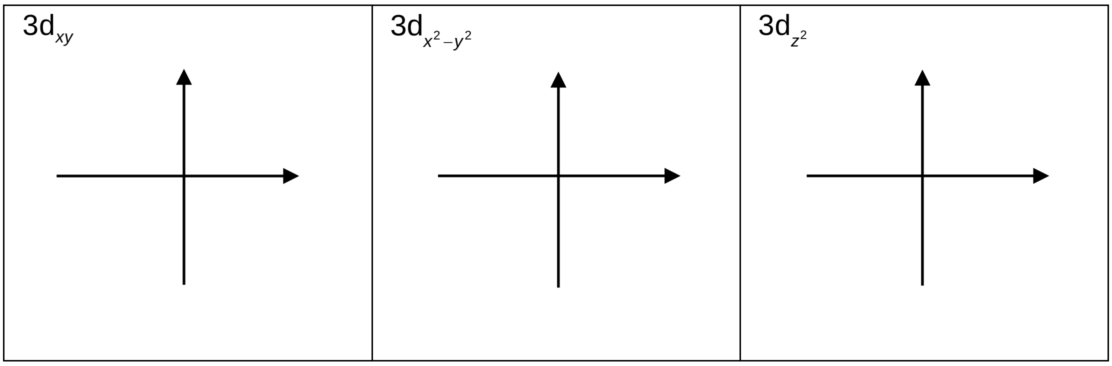
Cu^{+}: $1s^22s^22p^6$.. [1]

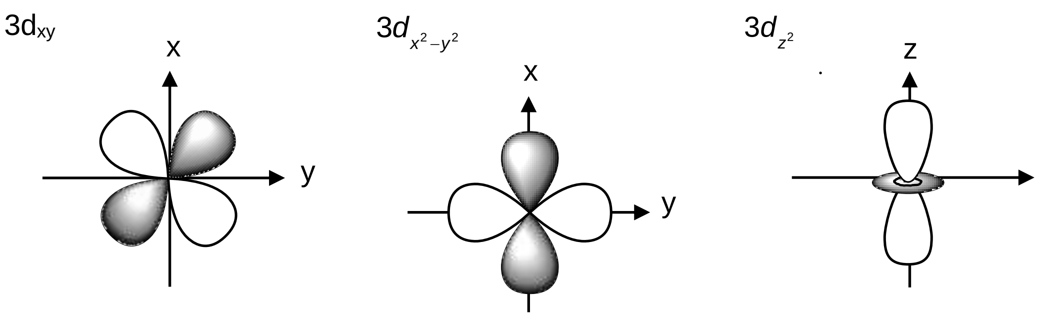
Cu^{+}: $1s^22s^22p^63s^23p^63d^{10}$

Comments

- Write the configuration for Cu first. This is a special case where there is a stable fully filled 3d subshell configuration: **Cu: $1s^22s^22p^63s^23p^63d^{10}4s^1$**
- Once the 3d subshell contains electrons, the 4s will be higher in energy than 3d.
- Thus we need to write 3d below 4s and not ……$4s^13d^{10}$
- To form Cu^{+}, remove the 4s electron first.
- Thus **Cu^{+}: $1s^22s^22p^63s^23p^63d^{10}$**

(b) Sketch the $3d_{xy}$, $3d_{x^2-y^2}$ and $3d_{z^2}$ orbitals and label all appropriate axes. [2]





Features of correctly labelled orbitals
Shape of orbital between / along axis
Label appropriate axis

- between x and y for $3d_{xy}$
- along x and y for $3d_{x^2-y^2}$
- along z for $3d_{z^2}$

Comments

- All relevant axis must be labelled
- For $d_z{}^2$, the doughnut shaped lobe must be drawn.

For Examiner's Use

(c) Complete Table 2.1 to describe the structure and bonding for ethane-1,2-diol and copper.

Table 2.1

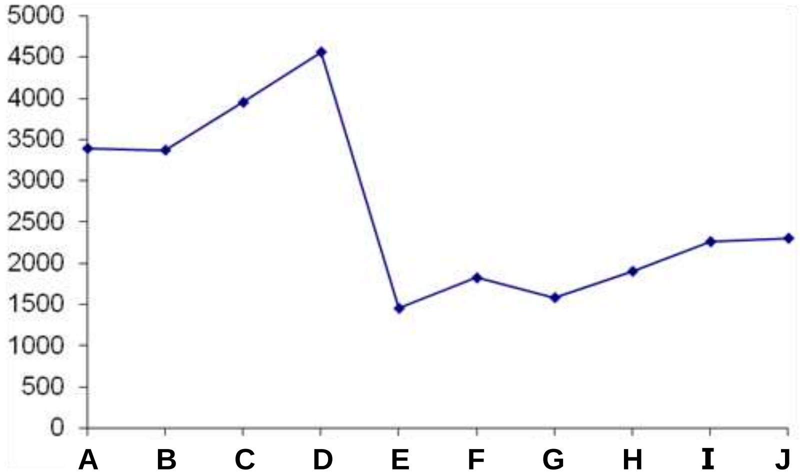
	ethane-1,2-diol	copper
structure		
bonding		
		[3]
structure	**simple molecular/covalent structure**	**giant metallic lattice structure**
bonding	**hydrogen bonding between molecules and covalent bonds between atoms**	**electrostatic forces of attraction between cations and** sea of **delocalised electrons**

Comments

- The question asked for description of structure, not a diagram. Thus you should describe the structure (simple molecular, giant metallic lattice etc).
- For Cu structure, "giant" and "lattice" must be clearly indicated. This is also true for all other giant structures like giant ionic lattice or giant covalent lattice.
- For ethane-1,2-diol, a common mistake was not to state the interatomic covalent bonds between the atoms.
- It is insufficient to just state hydrogen bonding for ethane-1,2-diol. This answer is unclear as you need to recognise that they hydrogen bonds are intermolecular forces **between ethane-1,2-diol molecules**.
- A common mistake was to state that hydrogen bonds is between O and H of ethane-1,2-diol. This shows a confusion that hydrogen bond is interatomic rather than intermolecular.
- You need to spell out "hydrogen bonds" at least once and not just use H-bond.
- For Cu, it is important to state that metallic bonds are between cations and delocalised electrons. A common mistake was to state atoms instead of cations.
- The term <u>delocalised electrons</u> is characteristic in metals and must be stated. Mobile or free moving electrons are not acceptable as these terms are to illustrate electrical conductivity rather than illustrate bonding.

[Total: 6]

For Examiner's Use

3		In an experiment, samples of elements **A** to **J**, with consecutive atomic numbers less than 20, were analysed and their **3rd** ionisation energies plotted as shown. **3rd I.E. / kJ mol^{-1}**  **Elements**
	(a)	**G** is a Group 15 element and can form chlorides with the formula **G**C*l*3 and **G**C*l*5. State the electronic configuration of the element **G**. [1]
		1s^2 2s^2 2p^6 3s^2 3p^3
		Comments • For group 15, there are 5 valence electrons and these are the sum of the electrons in the 3rd valance shell, both 3s and 3p, not only 3p. A common mistake was to put the configuration as …..3p^5.
	(b)	Write an equation for the 3rd ionisation energy of **G**. [1] **G^{2+} (g) → G^{3+}(g) + e^- or P^{2+} (g) → P^{3+}(g) + e^-**
		Comments • The state symbol (g) is a must in the answer. • Common mistake was G(g) → G^{3+}(g) + 3e. This is incorrect as the question asked for 3rd IE, thus you need to start from the +2 cation.
	(c)	Explain why the 3rd ionisation energies increase from element **B** to **D**. [2]
		B^{2+} 2s^2 2p^4 C^{2+} 2s^2 2p^5 D^{2+} 2s^2 2p^6 **From B^{2+}** to **D^{2+}**, **<u>nuclear charge increases</u> due to increase in proton number but <u>shielding effect is relatively constant</u> since the inner shell of electrons remain the same.** **There is <u>stronger nuclear attraction</u> between the nucleus and the valence electrons.** **More energy is required to remove the valence electron.**

For Examiner's Use

		Comments • This question is asking you to compare the 3rd IE from B to D, NOT 3 consecutive IE (1st, 2nd then 3rd IE) in each element. Hence you should not be discussing about the same number of protons attracting less electrons as electrons are being removed. • For 3rd IE, you are removing the 3rd electron from the +2 cation. • When comparing 3rd IE across period, need to mention **nuclear charge**, **shielding effect.** After which, comment on the effective nuclear charge and link it to **nuclear attraction.** • Note that effective nuclear charge (is the net result of nuclear charge and shielding effect) cannot be used as a term to replace nuclear attraction (which is the attractive force of the nucleus on valence electrons which eventually links to energy required to remove electron).
	(d)	Explain the large difference in 3rd ionisation energies between **D** and **E**. [1] **The 3rd ionisation energy of D is much higher than E as its valence electron is <u>the inner quantum shell and experiences greater nuclear attraction</u>. More energy is needed to remove the electrons in the inner quantum shell.**
		Comments • When comparing IE for elements in different periods, need to state that the valence electron is in an **inner** or **outer shell.** • The term "different shell" or "another shell" is ambiguous and not acceptable as it does not clearly indicate distance from nucleus. • After which, link your answer to **nuclear attraction.**
		[Total: 5]

For Examiner's Use

4 (a) The data of two alkanes are given below:

compound	M_r	boiling point / ^oC
hexane	86.0	68
2,3-dimethylbutane	86.0	58

Explain why the boiling point of 2,3-dimethylbutane differs from that of hexane. [2]

Both 2,3-dimethylbutane and hexane are isomer. However, 2,3-dimethylbutane has a spherical electron cloud while hexane has an elongated electron cloud.

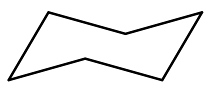
Hence, 2,3-dimethylbutane has a <u>smaller surface area of contact</u> with its other molecules and forms <u>weaker instantaneous dipole – induced dipole interactions</u>.
<u>Less amount of energy</u> is needed to overcome the weaker id-id interactions and thus 2,3-dimethylbutane has a lower boiling point.

Comments

- Note that both molecules have the same M_r hence they do not differ in electron cloud size. For similar M_r molecules with only id-id forces, the 2nd factor to look out for is branched vs straight chain and surface area of contact.
- You need to spell out **" instantaneous dipole – induced dipole interactions"** at least once and cannot just write id-id straightaway.

(b) Molecules will try to adopt the most stable *conformation* that minimises strain. *Conformation* refers to the shape that a molecule can adopt due to rotation around one or more single bonds.

For example, cyclohexane, C_6H_{12}, would adopt the following chair conformation instead of a planar hexagonal structure to minimise strain.



chair conformation

(i) Given that the hybridisation of carbon in cyclohexane is sp^3, state the bond angle of the C–C–C bond in the above chair conformation. [1]

109.5°

Comments:

- Generally well answered. Acceptable range from 109-110°

(ii) Draw all the hybrid orbitals of one of the carbon atoms in cyclohexane, showing the orientation of the hybrid orbitals clearly. [1]

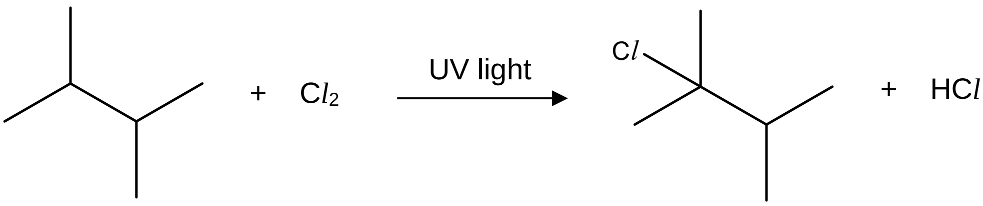
Comments:
- The 4 lobes should be drawn in a tetrahedral arrangement clearly and the small lobes must also be seen.
- Square planar shapes were not acceptable.
- X, Y and Z axis are not needed for this answer.
- A common mistake was to draw the 4 orbitals seperately. The question asked for the hybrid orbitals, not to show the process of hybridisation.

(iii) By considering your answer in **(b)(i)**, suggest why cyclopropane is not as stable as cyclohexane. [1]

Cyclopropane experiences considerable <u>angle strain</u> (60^o)/ <u>ring strain</u> as it <u>deviates from the ideal bond angle</u> (109.5^o).

Comments
- Since the question referred you to (b)(i), the focus must be on angle. Hence, approach the question based on bond angles in cyclopropane and the 109.5^o sp^3 angle you gave in (b)(i).
- Cyclopropane C atoms are sp^3 like cyclohexane C atoms. The interior angle of cyclopropane is 60^o and this deviates from the ideal sp^3 angle of 109.5^o thus the bond pairs will be closer together leading to significant bond pair bond pair repulsion.

(c) 2,3-dimethylbutane reacts in the following reaction to form 2-chloro-2,3-dimethylbutane.



2,3-dimethylbutane 2-chloro-2,3-dimethylbutane

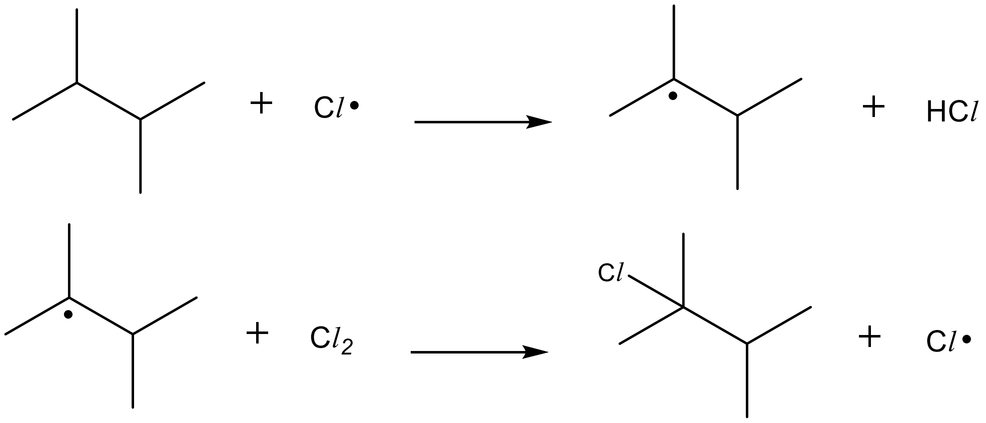
Name and outline the mechanism of the reaction between 2,3-dimethylbutane and chlorine to form 2-chloro-2,3-dimethylbutane. [4]

Free radical substitution

Initiation:

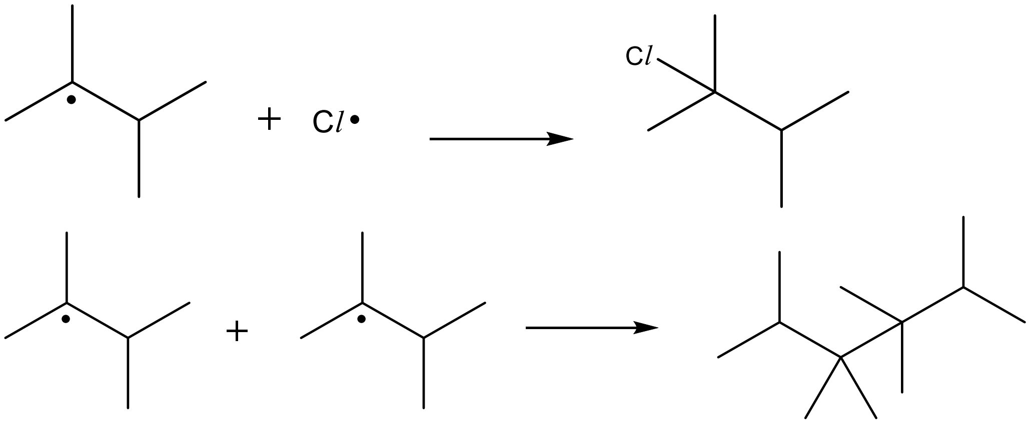
$$\mathbf{C}\boldsymbol{l} - \mathbf{C}\boldsymbol{l} \xrightarrow{\textbf{uv light}} \mathbf{2\ C}\boldsymbol{l}\bullet$$

Propagation:



Termination:

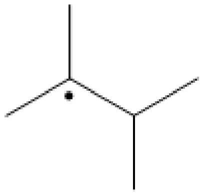
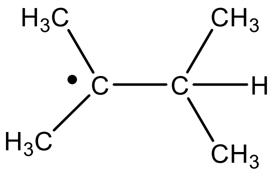
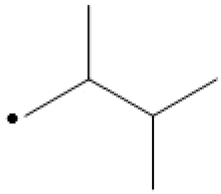
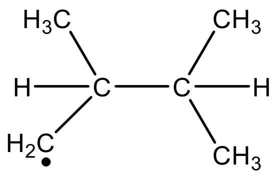
2 Cl• $\longrightarrow$ Cl$_2$

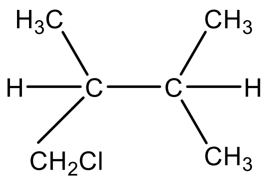


Check for the following:

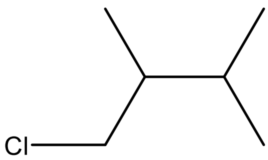
- ❖ Name of the three stages
- ❖ Conditions for initiation (uv light)
- ❖ Half arrows from bond to chlorine atoms in initiation step
- ❖ At least 2 termination steps to be given unless otherwise stated, and it should include the step which forms the required product
- ❖ Placing of • on the correct atoms in all steps

Comments:

- The correct radical to form the required product is  where the structure is: 
- A common mistake was to write the radical as  which is actually 
- Thus, your product will not be the given product and will be



Structural



skeletal

[Total: 9]

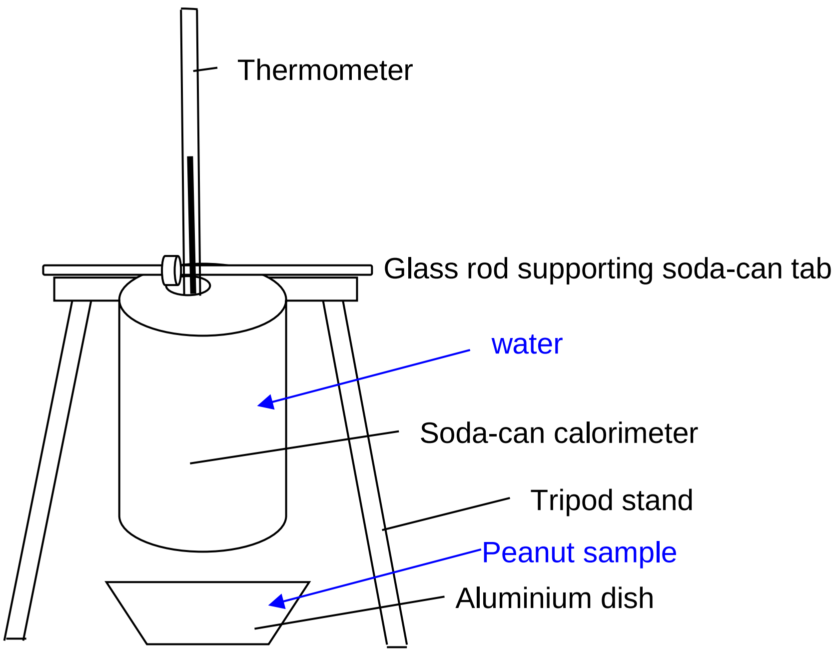
For Examiner's Use

5 The digestion and metabolism of food, such as peanuts in the body is similar to the combustion of fuels into carbon dioxide and water. In the process, energy is released. The total amount of energy released by the digestion and metabolism of a particular food is referred to as its *caloric content* and is expressed in terms of kcal which is usually shown on a food package.

The caloric content of a sample of peanuts was determined by conducting an experiment as given below.

Experiment procedure

1. The initial mass of a sample of peanuts was measured using a mass balance.
2. The initial temperature of 50 cm^3 of water was recorded using a thermometer.



3. The experiment was set up as shown in the diagram above.
4. The peanut sample was then ignited and allowed to burn for one minute.
5. The flame was then extinguished and the maximum temperature of the water was recorded.
6. The peanut sample was then cooled and reweighed.

(a) Draw two labelled arrows on the diagram to indicate where the peanut sample in step 4 and the 50 cm^3 of water in step 2 should be placed. [2]

Comments:

- The water should be placed in the soda can as that is where the thermometer is placed to monitor the temperature change of the water.
- The peanut sample is to be ignited so should be in the dish so that the heat produced from the combustion is used to heat the can of water. The dish of peanuts to be burnt is just like the ethanol spirit lamp placed at the bottom of the beaker of water.

For Examiner's Use

	(b)	Use the following data and the specific heat capacity of water from the *Data Booklet* to calculate the energy content for the peanut sample in J g^{-1}, given that [2]

Initial mass of peanut sample /g	4.650
Final mass of peanut sample /g	4.530
Initial temperature of water /^{o}C	29.0
Final temperature of water /^{o}C	36.0

Heat evolved when food sample is burnt = Heat gained by water
= (50 x 4.18 x 7.0)
= 1463 J

Mass of peanut sample burnt = 4.650 – 4.530
= 0.120g

Caloric content of food sample = Calories produced divided by mass of sample burnt
= 1463/0.120
= 12200J g^{-1}

Comments:

- A common mistake was to use the mass of peanuts as the m in mcΔT.
- The temperature change is that of water. So when using the formula mcΔT, the m should be the mass of the 50cm^3 of water used and since density is 1gcm^{-3}, the mass of water is 50g.
- The unit for q when using mcΔT is J since the unit for c is 4.18$Jg^{-1}K^{-1}$
- Since it is assumed that the heat absorbed by the water is equal to the heat released from the combustion of the given mass of peanuts, dividing q by the mass of peanuts burnt gives the energy content.
- The question mentioned that the energy content for the peanut sample in J g^{-1} hence this unit should be strictly followed.

(c) The actual caloric content of peanuts is 25500 J g^{-1}.

Suggest a reason for the discrepancy between your answer in (i) and the actual caloric content. [1]

Heat loss to surroundings OR Incomplete combustion of food sample.
OR The heat capacity of the calorimeter was not accounted for. Some heat was absorbed by the calorimeter.

Comments:

- You should clearly state that heat loss to the surroundings or incomplete combustion leads to less heat transferred to the soda can of water thus the ΔT is lesser leading to a lesser calculated q, hence the calculated energy content is less.
- It is incorrect to say that the heat capacity is assumed to be that of pure water. This is because in this case, pure water is placed in the can and the heat capacity would be 4.18$Jg^{-1}K^{-1}$

[Total: 5]

For Examiner's Use

Section C: Essay Questions (15 marks)

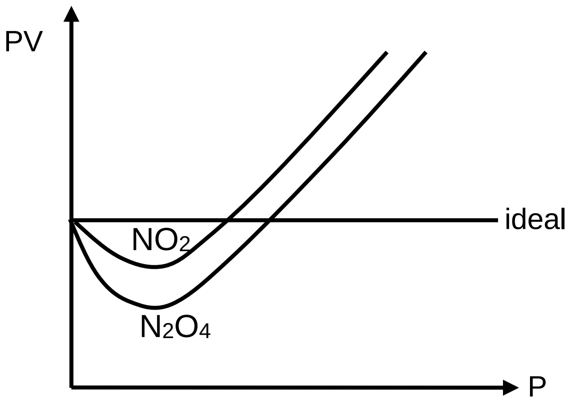
Answer **all** the questions from this section.

1 (a) Dinitrogen tetraoxide, N_2O_4, and nitrogen dioxide, NO_2, exist in dynamic equilibrium with each other as shown below.

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ ----------- **Equation 1**

The structures of N_2O_4 and NO_2 are found to be as follows.

The graphs of variation of PV against P at constant temperature for 1 mol N_2O_4 and 1 mol NO_2 are shown below.



Explain why N_2O_4 deviates more from ideality compared to NO_2. [1]

<u>N_2O_4 has a larger and more polarisable electron cloud</u> than NO_2. Hence <u>it has stronger instantaneous dipole- induced dipole interactions</u> than NO_2. The stronger id-id interactions of N_2O_4 lead to greater compressibility, hence it shows greater decrease in pV when P increases.

Comments

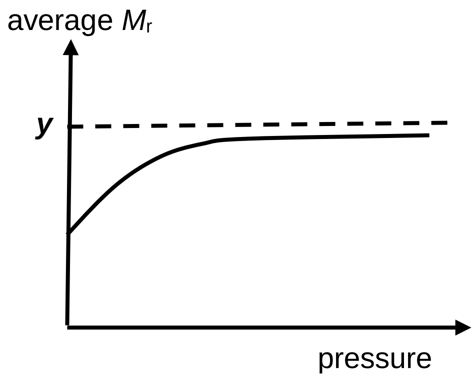
- The graph shows a more negative deviation for N_2O_4 hence the main reason should be the more significant intermolecular forces for N_2O_4.
- From the diagram of N_2O_4 given, you should be able to tell that the dipole moments of the four nitrogen-oxygen bonds cancel out thus the intermolecular forces should be instantaneous dipole-induced dipole.

(b) (i) Suggest the effect of increasing pressure on the position of the equilibrium in **Equation 1**. [1]

By Le Chatelier's Principle, as pressure increases, the <u>position of equilibrium would shift left to decrease the amount of gaseous molecules</u>.

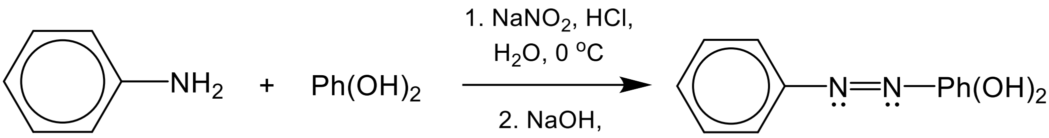
Comments

- It is not sufficient to just mention the position of equilibrium shifts to the left. You need to explain why the position of equilibrium shifts to the left by

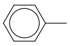
			mentioning that it is to partially offset the pressure increase by **decreasing the amount of gaseous molecules.**
		(ii)	The diagram below shows the variation of the average M_r of the equilibrium mixture with pressure for **Equation 1**, where average M_r is dependent on the relative proportions of N_2O_4 and NO_2.  Account for the change in average M_r of the mixture as pressure increases. Hence state the value of ***y***. [2]
			Since POE shifts to the left with an increase of pressure, more N_2O_4 will be produced and average M_r increases. ***y* = 2(14.0) + 4(16.0) = 92.0 (1d.p.)**
			Comments • The keywords in the question is to account for the change in M_r and he value of y. • Since we observe that the average M_r increases, the higher M_r N_2O_4 must have been favoured and this must be clearly stated. • Since the M_r of N_2O_4 is 92.0, y is 92.0. Since the reaction still remains at equilibrium, there will always be some N_2O_4 dissociating back to NO_2, thus the M_r will only approach 92.0 and will never be equal to 92.0 since pure N_2O_4 in the system cannot be achieved.
		(iii)	Predict with reasoning, whether the sign of enthalpy change of the forward reaction is positive or negative. Hence suggest how the position of equilibrium will shift when temperature is increased. [2]
			The sign of enthalpy change of the forward reaction is positive as it involves the breaking of the N-N bond. OR backward rxn is exothermic, bond formation. **POE shifts to the right to absorb heat.**
			Comments • The question ask for the sign of the enthalpy change, hence your answer should be **positive** rather than endothermic as the word "endothermic" does not reflect the sign. • Answers which state energy required in bond breaking exceeds energy released in bond forming is not accepted since no bonds are formed at all. (Only N-N bond of N_2O_4 is broken to form $2NO_2$)

			• Once again, it is insufficient to just mention that POE shifts right. You need to explain why POE shifts right by stating that it is to absorb heat to partially offset the increase in temperature.
	(c)	**(i)**	0.0500 mol of N_2O_4 was injected into a sealed 1dm^3 vessel containing 0.0100 mol of inert neon gas. When equilibrium was established at 50 ^oC, the total pressure was 1.95 bar. Determine the total amount of gases at equilibrium. [2]
			$(1.95 \times 10^5)(10^{-3})$ = n (8.31)(273+50) [1] converting the units of P and T **n = 0.0726 mol**
			Comments: • The pressure in bar should be converted to Pa by multiplying by 10^5 since 1 bar = 100 000 Pa. This conversion is in the Data Booklet. • The volume 1dm^3 should be converted to m^3 by dividing by 1000. • Temperature should be converted to K. • Unit should be given in mol
		(ii)	Determine the partial pressures of N_2O_4 and NO_2 at equilibrium. [2]
			$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ **Initial / mol: 0.0500, 0** **Change: -x, +2x** **Equilibrium / mol: 0.0500-x, 2x** **$\underbrace{0.0500-x}_{N_2O_4} + \underbrace{2x}_{NO_2} + \underbrace{0.0100}_{Ne} = 0.0726$** **x = 0.01265 mol** **$P_{N2O4} = \frac{0.0500-0.01265}{0.07265} \times 1.95 = 1.003 = 1.00$ bar** **$P_{NO2} = \frac{2\times0.01265}{0.07265} \times 1.95 = 0.6791 = 0.679$ bar**
			Comments: • The amount of gas 0.0726 mol includes N_2O_4, NO_2 and 0.0100 mol Ne. • Note that Ne does not take part in the equilibrium since the question already mentioned Ne is an inert gas. • A common mistake was to take **0.0500-x + 2x = 0.0726** where the 0.0100mol of Ne was forgotten. • The answer for partial pressures should be kept in bar to avoid unnecessary conversions. You are not required to convert your answer to Pa or atm.
		(iii)	Hence calculate the value of K_P. [1]
			$K_P = \frac{0.6791^2}{1.003} = 0.460$ bar
			Comments: • Generally well done. Only a small handful forgot to square the **P_{NO2}.**

(d) **(i)** Another nitrogen compound, sodium nitrite $NaNO_2$, is used in the synthesis azo compounds, a class of organic compounds that have been used as dyes.



Azo dye

(where Ph = )

Scheme 1

Azo compounds are known to exhibit cis-trans isomerism. This is due to the lone pairs of electrons being a distinct group on the N atom. With reference to the N=N double bond of the azo dye, explain why it shows cis-trans isomerism. [2]

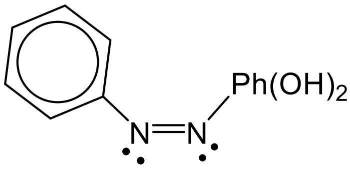
1. **Restricted rotation about the N=N double bond.**
2. **There are two different groups attached to the N atoms of the N=N double bond.**

Comments

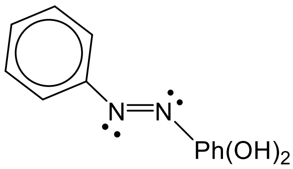
- It is insufficient to just state that there is an N=N double bond. It is important to mention that the double bond restricts free rotation.
- You should mention that there are two different groups on the same N of the N=N, not two different groups across two N atoms of the N=N bond.

(ii) Draw and label the cis and trans isomers based on your answer in **(d)(i)**. [2]

Cis

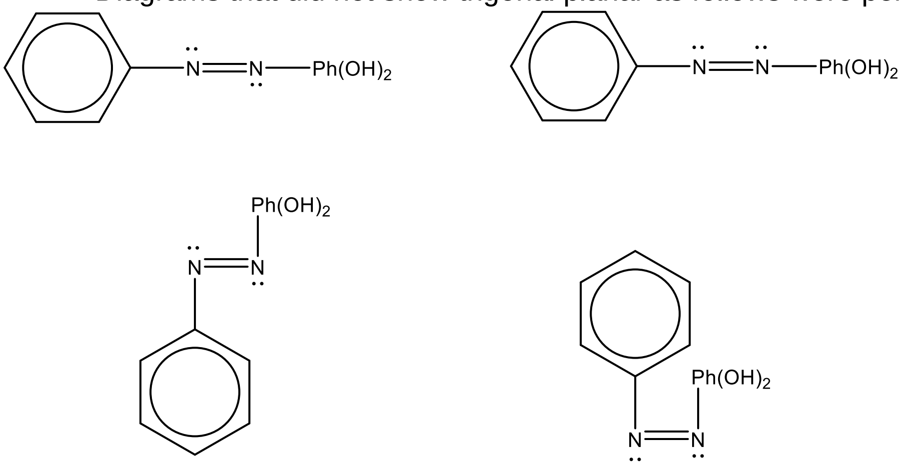


Trans



Comments

- For cis trans isomers, you need to clearly show the trigonal planar geometry at each N of the N=N double bond.
- Diagrams that did not show trigonal planar as follows were penalised:



[Total: 15]

~ End of paper ~

Section D: Essay Questions (15 marks)

Answer **one** question from this section.

1 (a) Methyl methanoate can be formed by reacting a mixture of carbon monoxide and methanol, using a solid sodium methoxide, CH_3ONa, catalyst at 70 °C.

$$CH_3OH(g) + CO(g) \xrightarrow{CH_3ONa} HCOOCH_3(g)$$

To determine the rate equation for this reaction, an investigation was carried at constant temperature and different partial pressures of CO, P_{CO}. The total pressure, P_T, of the system at the start of the reaction is also given.

experiment	**1**	**2**	**3**
initial rate / kPa min^{-1}	5.25	10.5	39.4
P_T / kPa	0.90	1.20	2.40
P_{CO} / kPa	0.30	0.60	0.90

(i) Determine the partial pressure of CH_3OH for experiment **1**, **2** and **3**. [1]

Pressure of CH_3OH for expt 1 = 0.9 – 0.3 = 0.600 kPa
expt 2 = 1.2 – 0.6 = 0.600 kPa
expt 3 = 2.4 – 0.9 = 1.50 kPa

Comments

- Answers should be given to 3 sig figures or 2 dec places as in the question with appropriate units.

(ii) Determine the order of reaction with respect to CH_3OH and CO. [2]

Using expt 1 and 2,

When the pressure of CO is doubled (pressure of CH_3OH remains constant), rate is doubled. Hence the order with respect to CO is 1.
(Can show the calculations)

Using expt 2 and 3,

$$\frac{39.4}{10.5} = \left(\frac{0.9}{0.6}\right)\left(\frac{1.5}{0.6}\right)^y$$

y = 1
Order with respect to CH_3OH is 1.

Comments

- The substitution method should be used to determine the order of CH_3OH.
- It is insufficient to just say that when using experiments 2 and 3, partial pressure of CO increase 1.5x and partial pressure of CH_3OH increase 2.5x so rate increase 3.75x. Hence first order for CH_3OH.
- You need to clearly mention that the 1.5x increase in partial pressure of CO caused an increase in 1.5x rate as CO is first order, we can attribute that 3.75/1.5 =2.5 x increase in rate is due to partial pressure of CH_3OH

increasing 2.5x, hence CH_3OH is first order, if you decide to use explanation instead of substitution.

(iii) Write the rate equation, in terms of partial pressure, for this reaction. [1]

Rate = k$P_{CH_3OH}P_{CO}$

Comments

- Rate equation must be written in terms of partial pressure.
- Square brackets should not be used as it represents concentration.
- The CH_3OH and CO should be subscript.

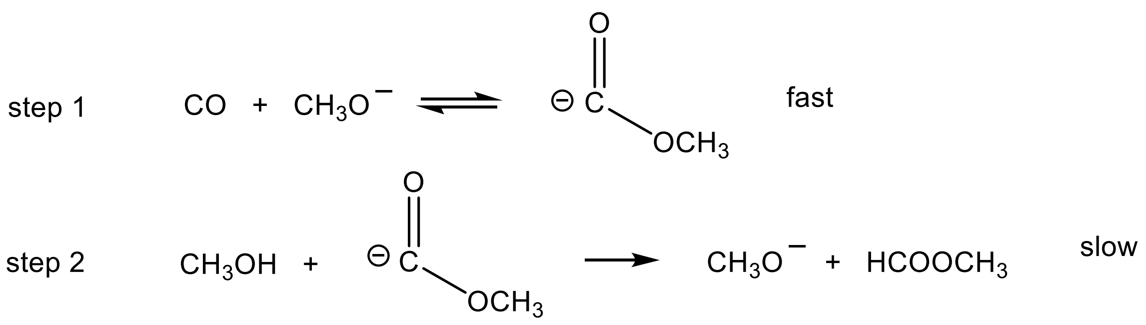
(iv) State one man-made source of carbon monoxide and its consequence. [2]

Incomplete combustion of fuel in car engines.
Decreases the oxygen carrying capability of blood / makes breathing more difficult

Comments

- Since CO is produced, it is important to mention that the combustion is *incomplete* to get credit. Combustion in fuels is insufficient.
- The immediate consequence that O_2 carrying capacity is reduced due to formation of CO binding to haemoglobin should be mentioned.

(b) A simplified reaction mechanism for the reaction is proposed as follows:



State whether CH_3O^- in step 1 is acting as an electrophile or nucleophile.
Explain your answer. [1]

Nucleophile. The lone pair of electrons on O is used to form a bond with the C of CO.

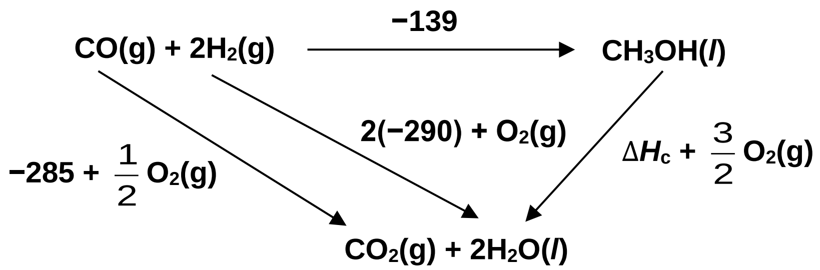
Comments

- The explanation should not just state that CH_3O^- is electron rich. You need to show how CH_3O^- behaves as a nucleophile in step 1, by stating that there is a donation of the **lone pair of electrons on O** to form a bond.
- Common mistakes include stating that an electron (or one electron) is donated instead of a pair of electrons. If only an electron is donated, then CH_3O^- would be a free radical rather than a nucleophile.

(c) The following data will be useful in this question.

equation	ΔH / kJ mol^{-1}
$CO(g) + 2H_2(g) \longrightarrow CH_3OH(l)$	− 139
$CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$	− 285
$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$	− 290

(i) By drawing an energy cycle, calculate the enthalpy change of combustion of methanol. [3]



$-285 + 2(-290) + 139 = \Delta H_c$
$\Delta H_c = -726$ kJ mol^{-1} (ECF)

Comments

- All state symbols must be given.
- Each arrow must be correctly balanced with the correct number of $O_2(g)$
- To ensure the $O_2(g)$ is balanced, the total number of $O_2(g)$ on the arrows in the same direction should equal the number of $O_2(g)$ on the arrows in the opposite directions.
- In this case, the 2 arrows in the same direction have ½ $O_2(g)$ + $O_2(g)$ added and this gives a total of 3/2 $O_2(g)$. This balances the 3/2 $O_2(g)$ added on the 3rd arrow that runs in the opposite direction.
- All arrows should also be clearly labelled with either the enthalpy change or the value of the enthalpy change, multiplied by the appropriate coefficients.
- For the enthalpy change, the – sign must be clearly shown. Units must be given and answers to 3 sig fig.

(ii) State the hybridisation of the following
- carbon atom in CO_2
- oxygen atom in H_2O [2]

C: sp **O: sp^3**

Comments

- C in CO_2 has 2 bond pairs and zero bond pairs, thus is sp
- O in H_2O has 3 bond pairs and 1 lone pair, thus 4 electron pairs in all. Hence H_2O is sp^3

(d) Consider the reaction:

		$$CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$$	
		(i)	Predict the sign of ΔS for the reaction. [1]
			ΔS is negative as there is a decrease in the number of gaseous particles, hence there are less number of ways to arrange the gaseous particles in the system/decreases in the disorderliness.
			Comments • It is important to state that the particles that decrease are the **gaseous particles** as change in number of gaseous particles causes the most significant change in entropy, disorder of the system.
		(ii)	Explain if the ΔG would be less or more exothermic compared to ΔH. [2]
			ΔG would be less exothermic compared to ΔH due to **$\Delta G = \Delta H - T\Delta S$** **(−) − (−)** **OR** **$-T\Delta S > 0$**
			Comments • You need to make use of the equation **$\Delta G = \Delta H - T\Delta S$** to comment how a negative ΔS affects $-T\Delta S$ and how this causes the ΔG to become less exothermic compared to ΔH
			[Total: 15]

2 Oxides of nitrogen refers to compounds that are composed of nitrogen and oxygen.

Some of the reactions involving the oxides of nitrogen and their standard enthalpy changes, $\Delta H^{\ominus}_{rxn}$ are given in the table below

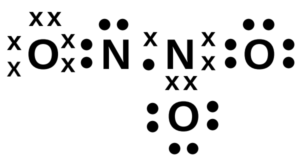
reaction	$\Delta H^{\ominus}_{rxn}$ / kJ mol^{-1}
(1) $N_2\,(g) + O_2\,(g) \longrightarrow 2NO\,(g)$	+ 180
(2) $NO\,(g) + ½\,O_2\,(g) \longrightarrow NO_2\,(g)$	– 57.0
(3) $NO\,(g) + NO_2\,(g) \longrightarrow N_2O_3\,(g)$	– 39.8
(4) $4NO\,(g) \longrightarrow N_2O\,(g) + N_2O_3\,(g)$	– 195

(a) State one man-made occurrence of oxides of nitrogen. [1]

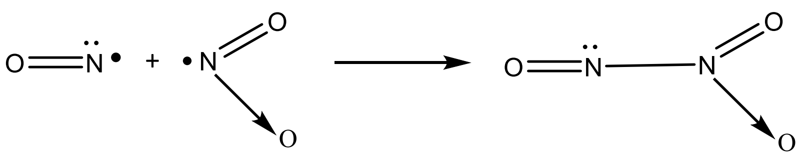
Produced from the reaction of N_2 with O_2 that takes place at a high temperature in the car engine / Produced from the combustion of fuels in internal combustion engines. / Produced from the burning of coal, oil, diesel fuel, and natural gas.

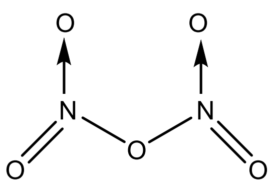
Comments
- By stating that N_2 and O_2 reacts is insufficient. The high temperature is required since N_2 has strong triple bonds between N atoms that needs to be broken.

(b) N_2O_3 is produced from NO and NO_2 as shown in **Reaction 3**. Given that there is an unpaired electron on both nitrogen atoms in NO and NO_2 respectively, draw the dot-and-cross diagram of N_2O_3. [1]



Comments
- You need to make use of the information given in the question to deduce the structure.
- Since there is an unpaired electron on both nitrogen atoms in NO and NO_2 respectively, we would expect a bond to form between the N atoms of NO and NO_2, thus there should be an N-N bond.



- The structure  was not accepted even though the octet configuration was fulfilled for all atoms, as it does not make use of the information in the question.

(c) Under conditions of high pressure and a catalyst, nitrogen monoxide, NO, forms two other oxides of nitrogen, dinitrogen monoxide, N_2O, and dinitrogen trioxide, N_2O_3, as shown in **reaction 4**.

(i) State clearly how the oxidation number of nitrogen changes during this reaction. [1]

Oxidation number of nitrogen decreases from +2 in NO to +1 in N_2O and increases to +3 in N_2O_3.

Comments

- You should clearly state whether the oxidation number decrease or increase and also clearly state the oxidation numbers in the respective species.
- Oxidation number should be written as +2, +1 etc and not 2+, 1+ etc.

(ii) Given that $\Delta G^\ominus$ of **reaction 4** is – 103 kJ mol^{-1}, calculate $\Delta S^\ominus$ in J mol^{-1} K^{-1} for this reaction at 298 K and explain the sign of $\Delta S^\ominus$. [2]

$\mathbf{\Delta S^\ominus = (\Delta H_{rxn} - \Delta G^\ominus) / T}$
$\mathbf{= (-195 + 103) / 298}$
$\mathbf{= -0.309\ kJ\ mol^{-1}\ K^{-1}}$
$\mathbf{= -309\ J\ mol^{-1}\ K^{-1}}$

$\Delta S^\ominus$ is negative as the <u>amount of gaseous particles decreases</u> from 4 mol to 2 mol, hence there are less number of ways to arrange the gaseous particles in the system/decreases in the disorderliness.

Comments

- The units has been given clearly in the question as J mol^{-1} K^{-1} thus you should give your answers corresponding to these units.
- It is important to state that the particles that decrease are the **gaseous particles** as change in number of gaseous particles causes the most significant change in entropy, disorder of the system.

(iii) Using relevant data in the table above, draw an energy cycle to calculate the standard enthalpy change of formation of N_2O_3. [3]

$$\begin{tikzcd} N_2(g) + {}^{3}/_{2}O_2(g) \arrow[r, "\Delta H_f^\ominus"] \arrow[d, "+180"'] & N_2O_3(g) \\ 2NO(g) + {}^{1}/_{2}O_2(g) \arrow[r, "-57.0"'] & NO(g) + NO_2(g) \arrow[u, "-39.8"'] \end{tikzcd}$$

$\mathbf{\Delta H_f^\ominus = (+180) + (-57.0) + (-39.8)}$
$\mathbf{= +83.2\ kJ\ mol^{-1}}$

Comments

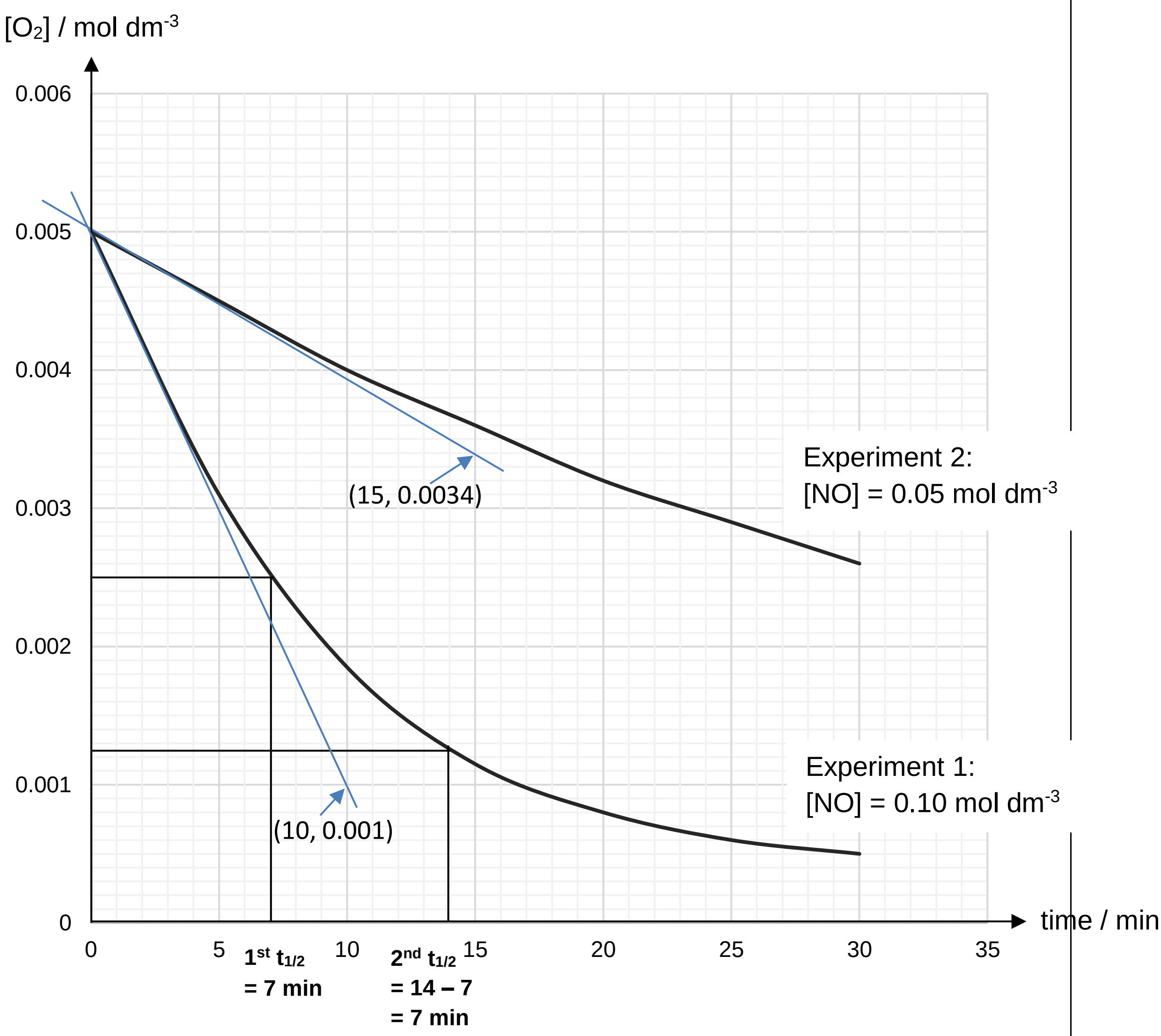
- All state symbols must be given.
- The $O_2(g)$ when one stage converts to the next should be balanced. A common mistake was the $O_2(g)$ was not balanced.
- All arrows should also be clearly labelled with either the enthalpy change or the value of the enthalpy change, multiplied by the appropriate coefficients.

- For the enthalpy change, the + sign must be clearly shown. Units must be given and answers to 3 sig fig.

(d) To study the kinetics of **reaction 2**, two experiments with different initial concentrations of NO were carried out at constant temperature.

Reaction 2: NO (g) + ½ O_2 (g) $\longrightarrow$ NO_2(g)

Based on the experimental results obtained, the graphs of [O_2] against time for the two experiments were plotted on the same axes on the **INSERT**.



(i) Deduce the order of reaction with respect to NO and O_2, showing clear working on the graph on the **INSERT**. [4]

From experiment 1, the <u>half-lives are constant at 7 min</u>, hence the reaction is <u>first order with respect to O_2</u>.

For experiment 1,

Gradient of tangent at t=0 is $\frac{0.005-0.0034}{0-15}$ = **<u>1.07 x 10^{-4} mol dm^{-3}min^{-1}</u>**

			For experiment 2, Gradient of tangent at t=0 is $\frac{0.005-0.001}{0-10}$ = **<u>4.00 x 10^{-4} mol $dm^{-3}min^{-1}$</u>** **Comparing experiment 1 and 2, the magnitude of the gradient of the tangent at t = 0 min increases from <u>1.07 x 10^{-4} to 4.00 x 10^{-4}</u>.** **This means that when <u>[NO] is doubled, the initial rate increases by approximately 4 times.</u> Rate of reaction $\propto$ $[NO]^2$, hence the reaction is <u>second order with respect to NO</u>.** **Alternatively:** **Student can determine the time taken for both graphs to fall from e.g. 0.005 to 0.004 and compare the times taken for both.** **The <u>time taken for expt 1 and expt 2 are 2.5 and 10 mins</u> respectively. Since the time taken decrease by 4 times or <u>rate increases by 4 times</u> when [NO] doubles, the reaction is <u>second order with respect to NO.</u>**
			Comments • All construction lines must be clearly shown. • For half life, you need to clearly show both half lives and indicate the half life values on the x-axis.You should also clearly show that the 2nd half life is 14-7 = 7min • When concluding order as being first order for O_2, you need to clearly state that the two half lives are **constant**. • For the initial rates, clear tangents should be drawn at t=0 and coordinates used to calculate the gradients should be clearly stated. • Clear working should also be shown for calculating the gradient of each tangent with answers to 3 sig fig with units. • When concluding order as being second order for NO you need to clearly state that when [NO] increase 2x, rate increases 4x.
		(ii)	Hence, state the rate equation of **reaction 2** and calculate the value of the rate constant. [2] **rate = k$[O_2][NO]^2$** **Using initial rate method (expt 1):** **4.00 x 10^{-4} = k(0.005)$(0.1)^2$** **k = 8.00 mol^{-2} dm^6 min^{-1}** **or using initial rate method (expt 2):** **1.07 x 10^{-4} = k(0.005)$(0.05)^2$** **k = 8.56 mol^{-2} dm^6 min^{-1}** **or using half-life method:** **rate = k'$[O_2]$ where k' = k$[NO]^2$ ([NO] is present in excess)** **k' = ln2 / 7 = 0.09902 min^{-1}** **k = 0.09902 / $(0.1)^2$ = 9.90 mol^{-2} dm^6 min^{-1}**
			Comments • All working must be clearly shown with the appropriate experiment chosen, clear substitution of the rate and concentrations of NO and O_2. • The units for k should be clearly stated.

		(iii)	State how the half-life of oxygen will be affected when the concentration of nitrogen monoxide is doubled. [1] **rate = $k[O_2][NO]^2$** **rate = $k'[O_2]$ where $k' = k[NO]^2$ ([NO] is present in excess)** **$t_{1/2}$ = ln2 / k'** **= ln2 / $k[NO]^2$** **When [NO] is doubled, the half-life of O_2 will be <u>reduced by 4 times</u>.**
			Comments • The initial concentration of NO is more than 10x that of O_2 so NO is in excess. • The reaction is pseudo first order since NO is present in excess thus the [NO] is a constant. • $t_{1/2}$ = ln2 / k' where k' is k[NO], we conclude $t_{1/2}$ = ln2 / $k[NO]^2$ • Since [NO] increase 2x, $[NO]^2$ will increase 4x • Since $t_{1/2}$ is inversely proportional to $[NO]^2$, $t_{1/2}$ will decrease by 4 x
			[Total: 15]

~ End of paper ~